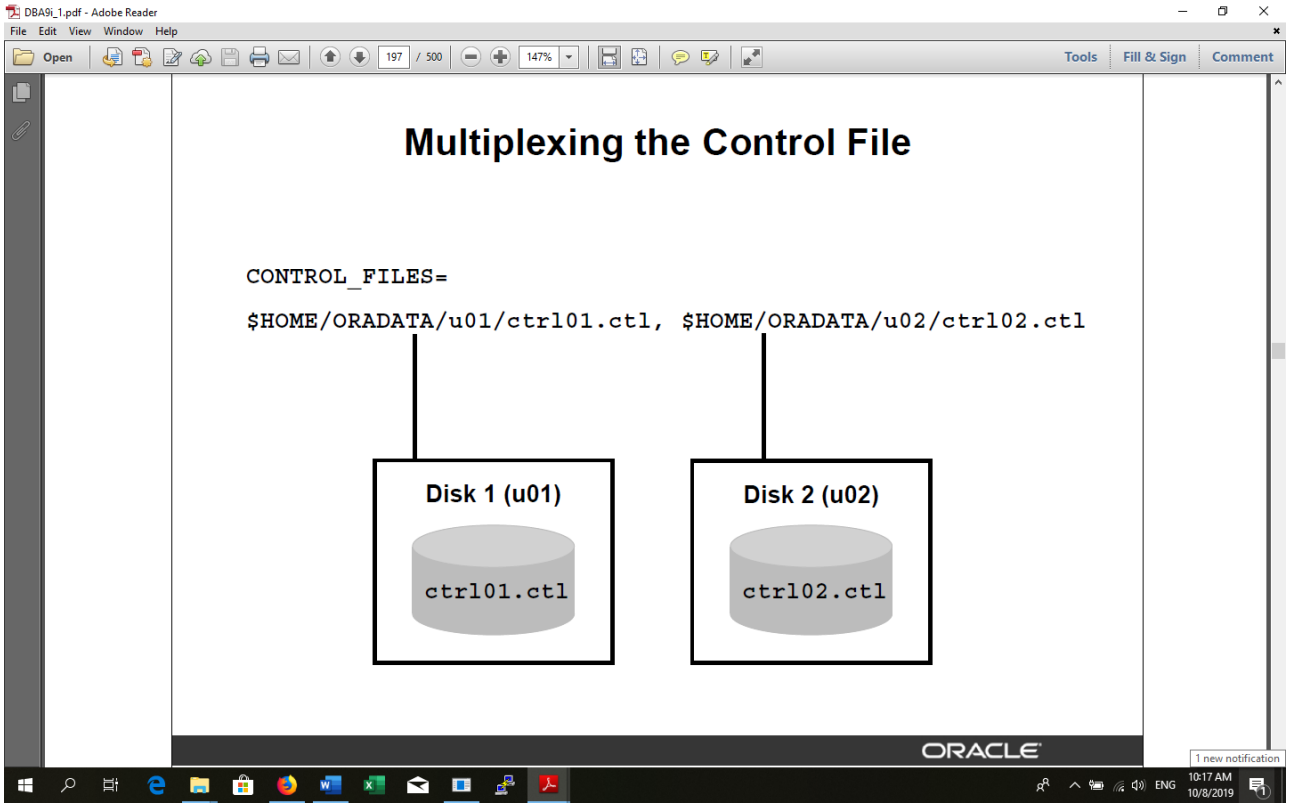
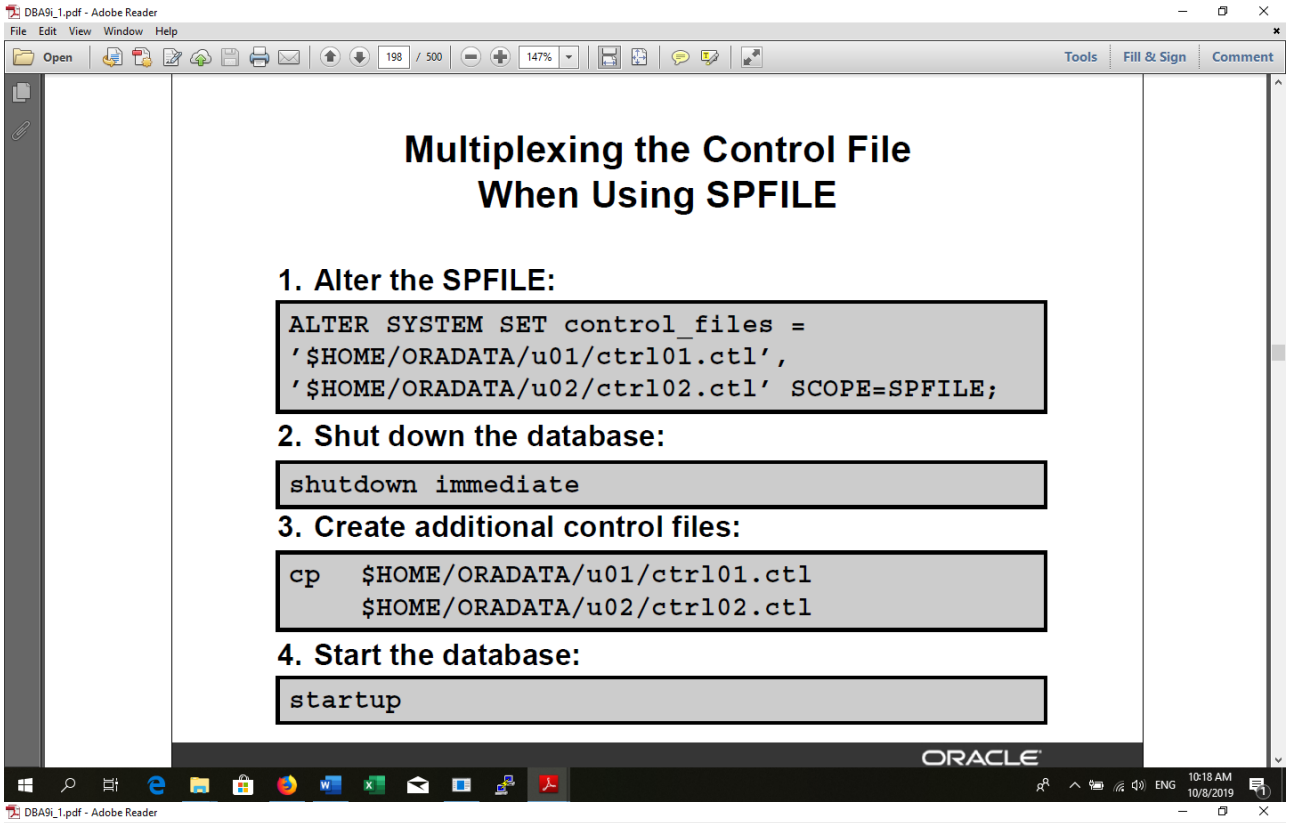
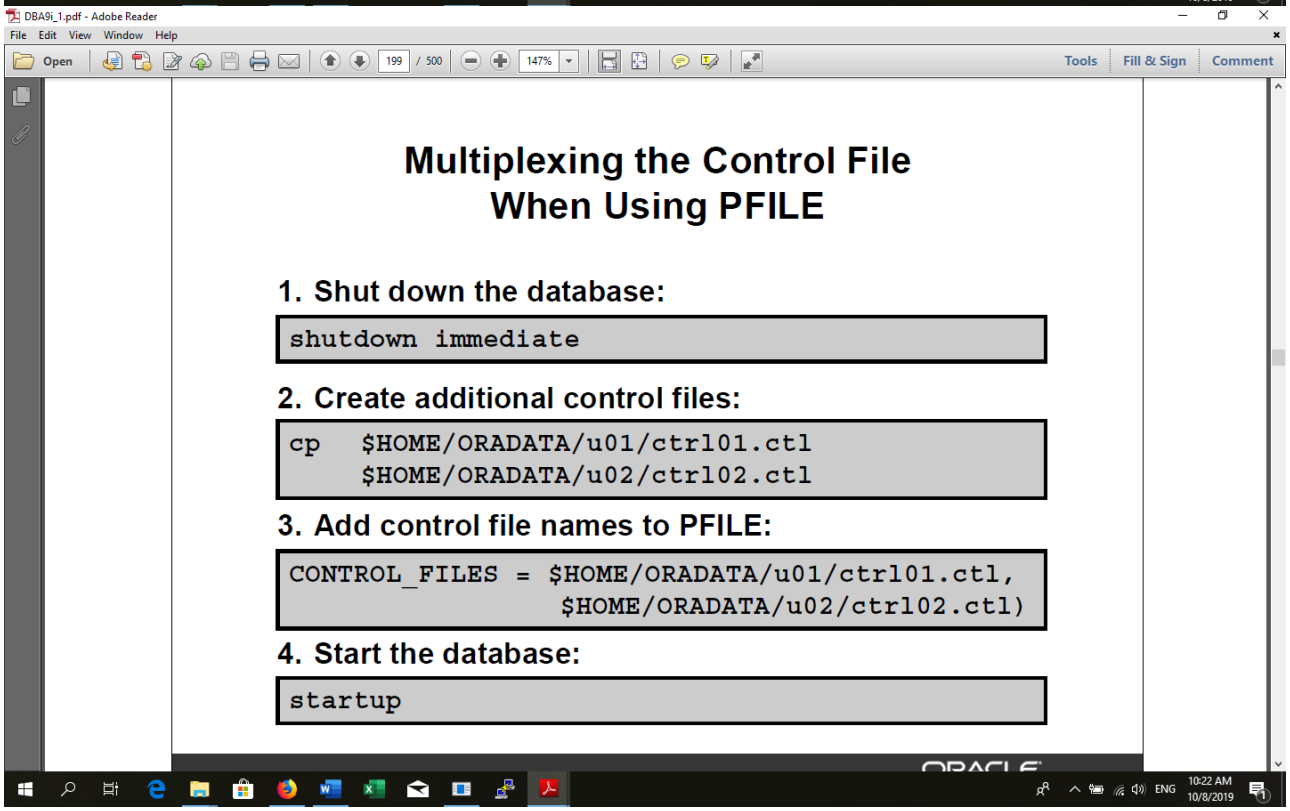
Fisiere de control si fisiere de log

A. Fisiere de control







Cateva exemple de view-uri dinamice care extrag informatii din fisierele de control:

• V$BACKUP
• V$DATAFILE
• V$TEMPFILE
• V$TABLESPACE
• V$ARCHIVE
• V$LOG
• V$LOGFILE
• V$LOGHIST
• V$ARCHIVED_LOG
• V$DATABASE

1) Informatii despre fisierele de control extrase din view-uri dinamice:

```
SQL> desc v$controlfile
Name                                                                                    Null?    Type
 ----------------------------------------------------------------------------------------------- ---------
 STATUS                                                                                          VARCHAR2(7)
 NAME                                                                                            VARCHAR2(513)
 IS_RECOVERY_DEST_FILE                                                                           VARCHAR2(3)
 BLOCK_SIZE                                                                                      NUMBER
 FILE_SIZE_BLKS                                                                                  NUMBER

SQL> select * from v$controlfile;

STATUS
-------
NAME
--------------------------------------------------
IS_     BLOCK_SIZE          FILE_SIZE_BLKS
---     ----------          --------------
F:\ORA10G_DB\PRODUCT\10.2.0\ORADATA\UPB\CONTROL01.CTL
NO       16384               430

F:\ORA10G_DB\PRODUCT\10.2.0\ORADATA\UPB\CONTROL02.CTL
NO       16384               430
```

2) Informatii despre fisierele de control extrase din view-ul pentru parametri:

```
SQL> desc v$parameter
 Name                          Null?    Type
 ----------------------------- -------- ----
 NUM                                    NUMBER
 NAME                                   VARCHAR2(64)
 TYPE                                   NUMBER
 VALUE                                  VARCHAR2(512)
 ISDEFAULT                              VARCHAR2(9)
```

```
ISSES_MODIFIABLE                 VARCHAR2(5)
ISSYS_MODIFIABLE                 VARCHAR2(9)
ISMODIFIED                       VARCHAR2(10)
ISADJUSTED                       VARCHAR2(5)
DESCRIPTION                      VARCHAR2(64)
UPDATE_COMMENT                   VARCHAR2(255)
```

```
SQL> select * from v$parameter where name='control_files';

    NUM NAME                                                 TYPE
--------- ------------------------------------------------------------ ---------
VALUE
--------------------------------------------------------------------------------------------------------
ISDEFAULT ISSES ISSYS_MOD ISMODIFIED ISADJ DESCRIPTION
--------- ----- --------- ---------- ----- ----------------------------------------------------------
UPDATE_COMMENT
--------------------------------------------------------------------------------------------------------
    219 control_files                                        2
g:\oracle\oradata\leu\CONTROL01.CTL, g:\oracle\oradata\leu\CONTROL02.CTL,
g:\oracle\oradata\leu\CONT
FALSE     FALSE FALSE     FALSE      FALSE control file names list
```

3) Informatii despre marimea inregistrarii, numarul total de inregistrari alocate si cele utilizate referitoare la parametrii de control:

```
SQL> desc v$controlfile_record_section


 Name                      Null?    Type
 -------------             ----------------- -------- ----
 TYPE                              VARCHAR2(20)
 RECORD_SIZE                       NUMBER
 RECORDS_TOTAL                      NUMBER
 RECORDS_USED                      NUMBER
 FIRST_INDEX                       NUMBER
 LAST_INDEX                        NUMBER
 LAST_RECID                        NUMBER
```

```
SQL> select * from v$controlfile_record_section;

TYPE                        RECORD_SIZE  RECORDS_TOTAL  RECORDS_USED  FIRST_INDEX  LAST_INDEX  LAST_RECID
--------------------------- -----------  -------------  ------------  -----------  ----------  ---------
DATABASE                        316             1             1            0           0           0
CKPT PROGRESS                   8180            11            0            0           0           0
REDO THREAD                     256             8             1            0           0           0
REDO LOG                        72              16            3            0           0           3
DATAFILE                        428             100           20           0           0           2025
FILENAME                        524             2298          26           0           0           0
TABLESPACE                      68              100           23           0           0           19
```

TEMPORARY FILENAME	56	100	3	0	0	3
RMAN CONFIGURATION	1108	50	0	0	0	0
LOG HISTORY	56	292	292	126	125	709
OFFLINE RANGE	200	163	0	0	0	0
ARCHIVED LOG	584	28	0	0	0	0
BACKUP SET	40	409	0	0	0	0
BACKUP PIECE	736	200	0	0	0	0
BACKUP DATAFILE	116	282	0	0	0	0
BACKUP REDOLOG	76	215	0	0	0	0
DATAFILE COPY	660	223	1	1	1	1
BACKUP CORRUPTION	44	371	0	0	0	0
COPY CORRUPTION	40	409	0	0	0	0
DELETED OBJECT	20	818	1	1	1	1
PROXY COPY	852	249	0	0	0	0
BACKUP SPFILE	36	454	0	0	0	0

4) Vizualizare fisiere de control:

```
SQL> SHOW PARAMETER CONTROL_FILES

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
control_files                        string      F:\ORA10G_DB\PRODUCT\10.2.0\OR
                                                 ADATA\UPB\CONTROL01.CTL, F:\OR
                                                 A10G_DB\PRODUCT\10.2.0\ORADATA
                                                 \UPB\CONTROL02.CTL, F:\ORA10G_
                                                 DB\PRODUCT\10.2.0\ORADATA\UPB\
                                                 CONTROL03.CTL
```

5) Informatii despre fisierele temporare:

```
SQL> desc v$tempfile

 Name                          Null?    Type
 ----------------------------- -------- ----
 FILE#                                  NUMBER
 CREATION_CHANGE#                       NUMBER
 CREATION_TIME                          DATE
 TS#                                    NUMBER
 RFILE#                                 NUMBER
 STATUS                                 VARCHAR2(7)
 ENABLED                                VARCHAR2(10)
 BYTES                                  NUMBER
 BLOCKS                                 NUMBER
 CREATE_BYTES                           NUMBER
 BLOCK_SIZE                             NUMBER
 NAME                                   VARCHAR2(513)
```

SQL> select * from v$tempfile;

```
    FILE# CREATION_CHANGE# CREATION_       TS#    RFILE# STATUS  ENABLED
BYTES    BLOCKS CREATE_BYTES BLOC
--------- ---------------- ---------          ---------          ---------          ------- ---------- --------- --------- -
---
NAME
-----------------------------------------------------------------------------------------------------
       1                0                2         1 ONLINE  READ WRITE  41943040      5120     41943040
8192
G:\ORACLE\ORADATA\LEU\TEMP01.DBF
```

6) Informatii despre tablespace-uri:

SQL> desc v$tablespace

Name	Null?	Type
TS#		NUMBER
NAME		VARCHAR2(30)
INCLUDED_IN_DATABASE_BACKUP		VARCHAR2(3)

SQL> select * from v$tablespace;

TS#	NAME	INC
3	CWMLITE	YES
4	DRSYS	YES
5	EXAMPLE	YES
6	INDX	YES
7	ODM	YES
0	SYSTEM	YES
8	TOOLS	YES
1	UNDOTBS1	YES
9	USERS	YES
10	XDB	YES
2	TEMP	YES

11 rows selected.

7) Informatii despre baza de date si fisierele de control:

SQL> desc v$database

SQL> desc v$database

Name	Null?	Type

DBID	NUMBER
NAME	VARCHAR2(9)
CREATED	DATE
RESETLOGS_CHANGE#	NUMBER
RESETLOGS_TIME	DATE
PRIOR_RESETLOGS_CHANGE#	NUMBER
PRIOR_RESETLOGS_TIME	DATE
LOG_MODE	VARCHAR2(12)
CHECKPOINT_CHANGE#	NUMBER
ARCHIVE_CHANGE#	NUMBER
CONTROLFILE_TYPE	VARCHAR2(7)
CONTROLFILE_CREATED	DATE
CONTROLFILE_SEQUENCE#	NUMBER
CONTROLFILE_CHANGE#	NUMBER
CONTROLFILE_TIME	DATE
OPEN_RESETLOGS	VARCHAR2(11)
VERSION_TIME	DATE
OPEN_MODE	VARCHAR2(20)
PROTECTION_MODE	VARCHAR2(20)
PROTECTION_LEVEL	VARCHAR2(20)
REMOTE_ARCHIVE	VARCHAR2(8)
ACTIVATION#	NUMBER
SWITCHOVER#	NUMBER
DATABASE_ROLE	VARCHAR2(16)
ARCHIVELOG_CHANGE#	NUMBER
ARCHIVELOG_COMPRESSION	VARCHAR2(8)
SWITCHOVER_STATUS	VARCHAR2(20)
DATAGUARD_BROKER	VARCHAR2(8)
GUARD_STATUS	VARCHAR2(7)
SUPPLEMENTAL_LOG_DATA_MIN	VARCHAR2(8)
SUPPLEMENTAL_LOG_DATA_PK	VARCHAR2(3)
SUPPLEMENTAL_LOG_DATA_UI	VARCHAR2(3)
FORCE_LOGGING	VARCHAR2(39)
PLATFORM_ID	NUMBER
PLATFORM_NAME	VARCHAR2(101)
RECOVERY_TARGET_INCARNATION#	NUMBER
LAST_OPEN_INCARNATION#	NUMBER
CURRENT_SCN	NUMBER
FLASHBACK_ON	VARCHAR2(18)
SUPPLEMENTAL_LOG_DATA_FK	VARCHAR2(3)
SUPPLEMENTAL_LOG_DATA_ALL	VARCHAR2(3)
DB_UNIQUE_NAME	VARCHAR2(30)
STANDBY_BECAME_PRIMARY_SCN	NUMBER
FS_FAILOVER_STATUS	VARCHAR2(22)
FS_FAILOVER_CURRENT_TARGET	VARCHAR2(30)
FS_FAILOVER_THRESHOLD	NUMBER

```
FS_FAILOVER_OBSERVER_PRESENT           VARCHAR2(7)
FS_FAILOVER_OBSERVER_HOST              VARCHAR2(512)
CONTROLFILE_CONVERTED                  VARCHAR2(3)
PRIMARY_DB_UNIQUE_NAME                 VARCHAR2(30)
SUPPLEMENTAL_LOG_DATA_PL               VARCHAR2(3)
MIN_REQUIRED_CAPTURE_CHANGE#           NUMBER
CDB                                    VARCHAR2(3)
CON_ID                                 NUMBER
PENDING_ROLE_CHANGE_TASKS              VARCHAR2(512)
CON_DBID                               NUMBER
FORCE_FULL_DB_CACHING                  VARCHAR2(3)
```

```
SQL> select controlfile_type, controlfile_sequence#, controlfile_change#, controlfile_time
       from v$database;

CONTROL    CONTROLFILE _SEQUENCE#   CONTROLFILE_CHANGE#  CONTROLFI
-------    ----------------------   -------------------  ---------
CURRENT              15142                4912527        08-OCT-19
```

```
SQL> select * from v$database;

    DBID NAME      CREATED   RESETLOGS_CHANGE# RESETLOGS
---------- --------- --------- ----------------- ---------
PRIOR_RESETLOGS_CHANGE# PRIOR_RES LOG_MODE     CHECKPOINT_CHANGE#
----------------------- --------- ---------- ------------------
ARCHIVE_CHANGE# CONTROL CONTROLFI CONTROLFILE_SEQUENCE# CONTROLFILE_CHANGE#
--------------- ------- --------- --------------------- -------------------
CONTROLFI OPEN_RESETL VERSION_T OPEN_MODE          PROTECTION_MODE
--------- ----------- --------- -------------------- --------------------
PROTECTION_LEVEL     REMOTE_A ACTIVATION# SWITCHOVER# DATABASE_ROLE
-------------------- -------- ----------- ----------- ----------------
ARCHIVELOG_CHANGE# ARCHIVEL SWITCHOVER_STATUS    DATAGUAR GUARD_S SUPPLEME SUP
------------------ -------- -------------------- -------- ------- -------- ---
SUP FORCE_LOGGING                      PLATFORM_ID
--- ---------------------------------------- -----------
PLATFORM_NAME
--------------------------------------------------------------------------------
RECOVERY_TARGET_INCARNATION# LAST_OPEN_INCARNATION# CURRENT_SCN
---------------------------- ---------------------- -----------
FLASHBACK_ON       SUP SUP DB_UNIQUE_NAME
------------------ --- --- ------------------------------
STANDBY_BECAME_PRIMARY_SCN FS_FAILOVER_STATUS     FS_FAILOVER_CURRENT_TARGET
-------------------------- ---------------------- ------------------------------
FS_FAILOVER_THRESHOLD FS_FAIL
--------------------- -------
FS_FAILOVER_OBSERVER_HOST
--------------------------------------------------------------------------------
CON PRIMARY_DB_UNIQUE_NAME         SUP MIN_REQUIRED_CAPTURE_CHANGE# CDB
```

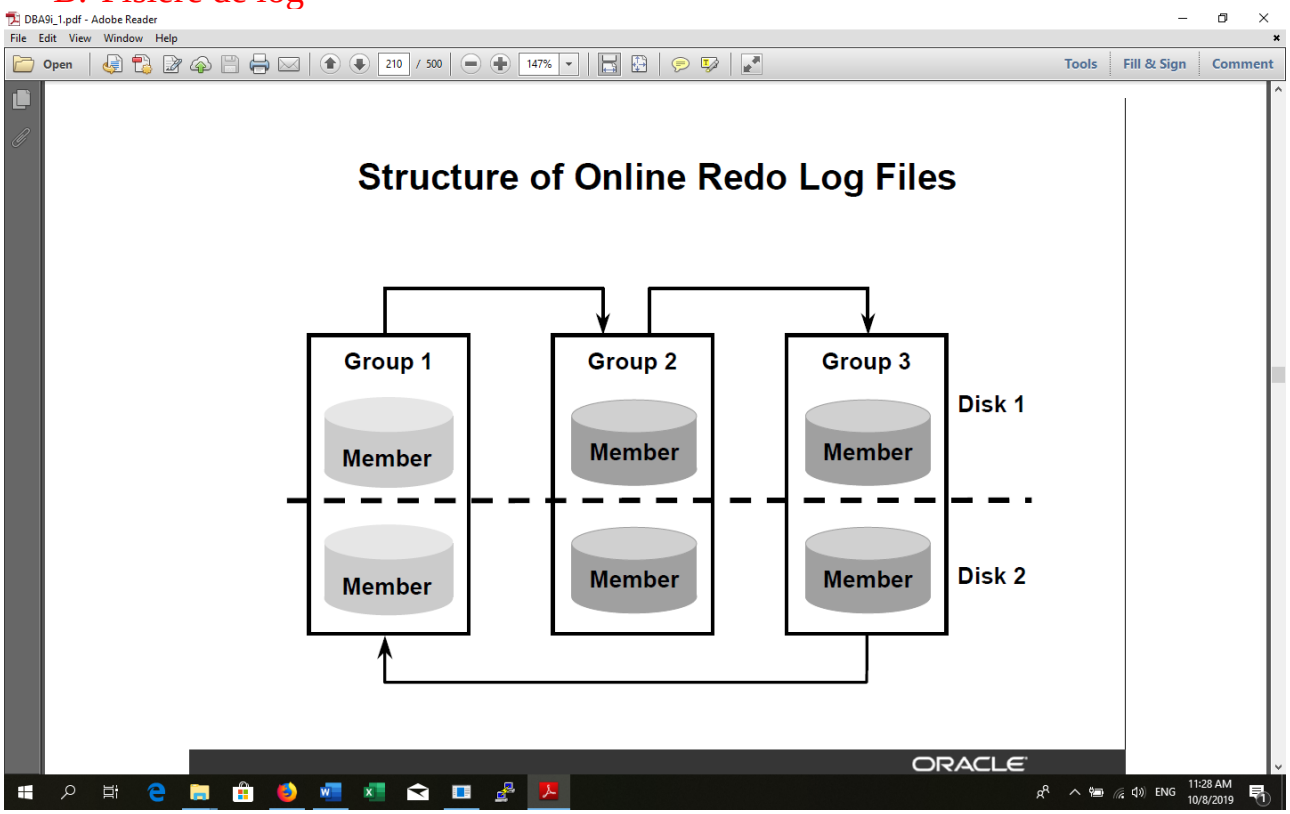
--- ------------------------------ --- -------------------------- ---
CON_ID

PENDING_ROLE_CHANGE_TASKS

CON_DBID FOR
---------- ---
1981261066 BD 08-MAY-19 2233668 08-MAY-19
1 11-SEP-14 NOARCHIVELOG 4867328
4822048 CURRENT 08-MAY-19 15148 4912912
08-OCT-19 NOT ALLOWED 08-MAY-19 READ WRITE MAXIMUM PERFORMANCE
UNPROTECTED ENABLED 1981264138 1981264138 PRIMARY
0 DISABLED NOT ALLOWED DISABLED NONE NO NO
NO NO 12
Microsoft Windows x86 64-bit
2 2 4912962
NO NO NO BD
0 DISABLED
0

NO NO NO
0
NOT APPLICABLE
1981261066 NO

B. Fisiere de log



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Forcing Log Switches and Checkpoints

- Forcing a log switch:

```
ALTER SYSTEM SWITCH LOGFILE;
```

- Checkpoints can be forced by:
  - Setting `FAST_START_MTTR_TARGET` parameter

```
FAST_START_MTTR_TARGET = 600
```

  - `ALTER SYSTEM CHECKPOINT` command

```
ALTER SYSTEM CHECKPOINT;
```

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Adding Online Redo Log File Groups

```
ALTER DATABASE ADD LOGFILE GROUP 3
('$HOME/ORADATA/u01/log3a.rdo',
 '$HOME/ORADATA/u02/log3b.rdo')
SIZE 1M;
```



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Adding Online Redo Log File Members

```
ALTER DATABASE ADD LOGFILE MEMBER
'$HOME/ORADATA/u04/log1c.rdo' TO GROUP 1,
'$HOME/ORADATA/u04/log2c.rdo' TO GROUP 2,
'$HOME/ORADATA/u04/log3c.rdo' TO GROUP 3;
```



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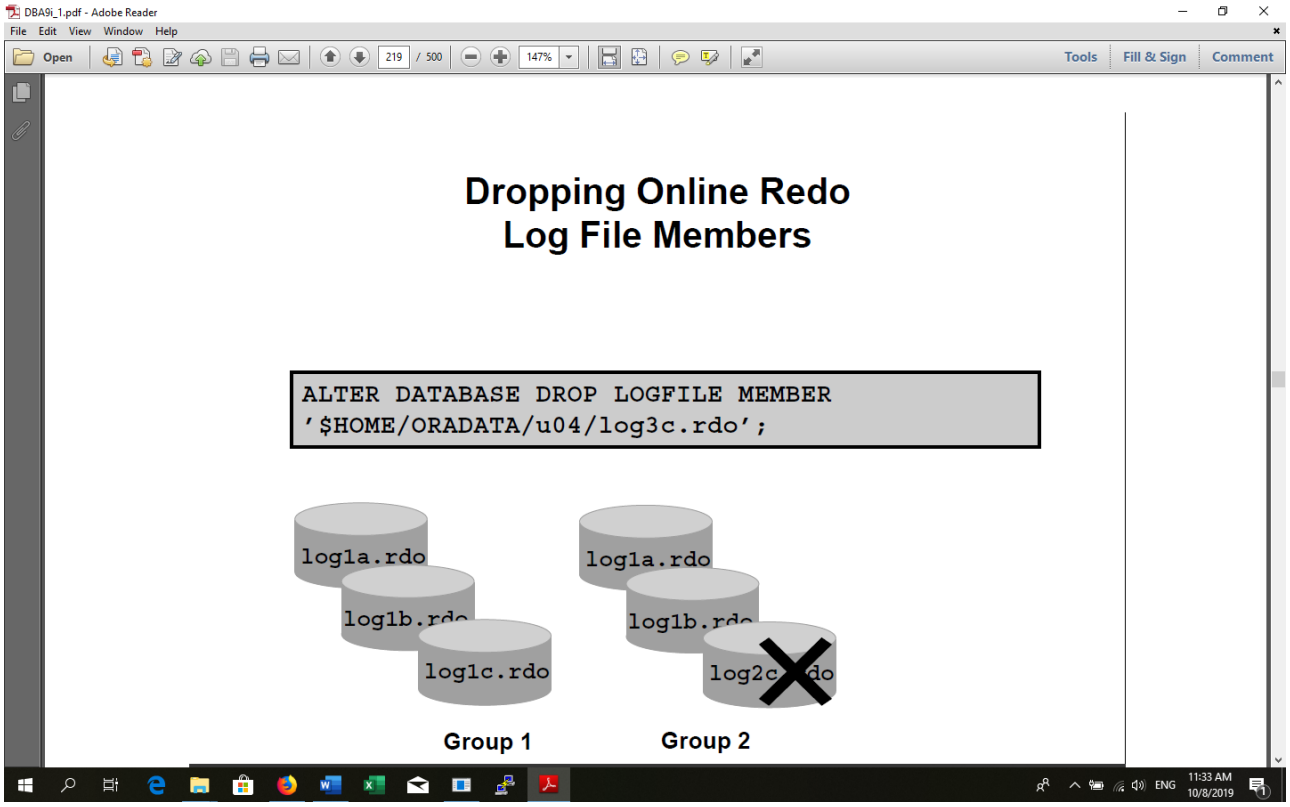
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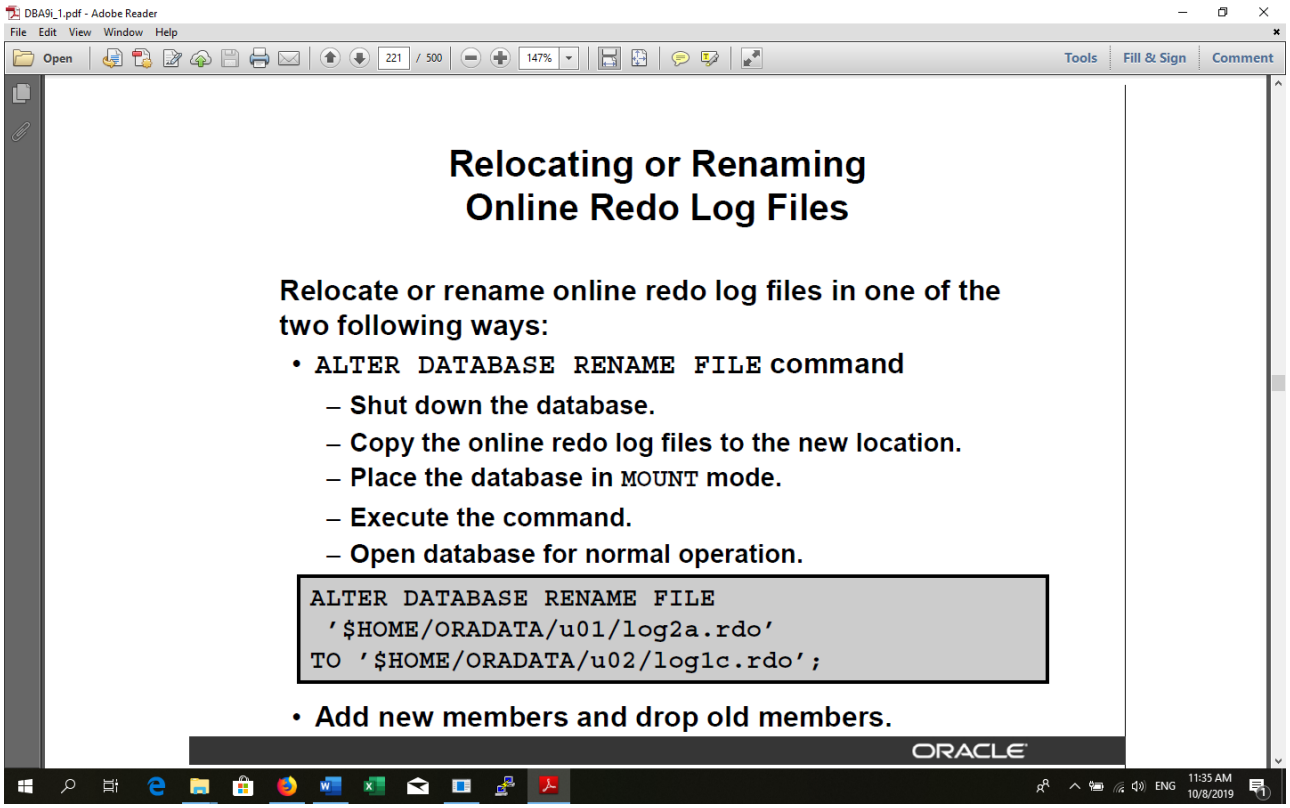
Dropping Online Redo Log File Groups

```
ALTER DATABASE DROP LOGFILE GROUP 3;
```



ENG 11:33 AM 10/8/2019





8) Informatii despre fisierele de log si starea lor:

SQL> desc v$logfile

Name	Null?	Type
GROUP#		NUMBER
STATUS		VARCHAR2(7)
TYPE		VARCHAR2(7)
MEMBER		VARCHAR2(513)

SQL> select * from v$logfile;

GROUP#	STATUS	TYPE	MEMBER	IS_
3		ONLINE	C:\ORACLE_12C\ORADATA\BD\REDO03.LOG	NO
2		ONLINE	C:\ORACLE_12C\ORADATA\BD\REDO02.LOG	NO
1		ONLINE	C:\ORACLE_12C\ORADATA\BD\REDO01.LOG	NO

9) Informatii legate de modul de lucru al bazei de date (cu arhivare sau fara arhivare a fisierelor de log, data cand au fost resetate fisierele de log, etc.)

SQL> desc v$database

Name	Null?	Type
DBID		NUMBER
NAME		VARCHAR2(9)
CREATED		DATE
RESETLOGS_CHANGE#		NUMBER
RESETLOGS_TIME		DATE
PRIOR_RESETLOGS_CHANGE#		NUMBER
PRIOR_RESETLOGS_TIME		DATE
LOG_MODE		VARCHAR2(12)
CHECKPOINT_CHANGE#		NUMBER
ARCHIVE_CHANGE#		NUMBER
CONTROLFILE_TYPE		VARCHAR2(7)
CONTROLFILE_CREATED		DATE
CONTROLFILE_SEQUENCE#		NUMBER
CONTROLFILE_CHANGE#		NUMBER
CONTROLFILE_TIME		DATE
OPEN_RESETLOGS		VARCHAR2(11)
VERSION_TIME		DATE
OPEN_MODE		VARCHAR2(10)
PROTECTION_MODE		VARCHAR2(20)

PROTECTION_LEVEL	VARCHAR2(20)
REMOTE_ARCHIVE	VARCHAR2(8)
ACTIVATION#	NUMBER
DATABASE_ROLE	VARCHAR2(16)
ARCHIVELOG_CHANGE#	NUMBER
SWITCHOVER_STATUS	VARCHAR2(18)
DATAGUARD_BROKER	VARCHAR2(8)
GUARD_STATUS	VARCHAR2(7)
SUPPLEMENTAL_LOG_DATA_MIN	VARCHAR2(3)
SUPPLEMENTAL_LOG_DATA_PK	VARCHAR2(3)
SUPPLEMENTAL_LOG_DATA_UI	VARCHAR2(3)
FORCE_LOGGING	VARCHAR2(3)

SQL> select name, log_mode, resetlogs_time from v$database;

NAME	LOG_MODE	RESETLOGS
UPB	NOARCHIVELOG	13-FEB-11

10) Informatii legate de starea instantei, a grupului current si a secventei curente

SQL> desc v$thread

Name	Null?	Type
THREAD#		NUMBER
STATUS		VARCHAR2(6)
ENABLED		VARCHAR2(8)
GROUPS		NUMBER
INSTANCE		VARCHAR2(16)
OPEN_TIME		DATE
CURRENT_GROUP#		NUMBER
SEQUENCE#		NUMBER
CHECKPOINT_CHANGE#		NUMBER
CHECKPOINT_TIME		DATE
ENABLE_CHANGE#		NUMBER
ENABLE_TIME		DATE
DISABLE_CHANGE#		NUMBER
DISABLE_TIME		DATE

SQL> select groups, current_group#, sequence#, instance, status from v$thread;

GROUPS	CURRENT_GROUP#	SEQUENCE#	INSTANCE	STATUS

3 1 872 upb OPEN

11) Informatii despre gupuri si membri in cadrul grupurilor:

```
SQL> desc v$log
 Name                          Null?    Type
 ----------------------------- --------  ----
 GROUP#                                  NUMBER
 THREAD#                                 NUMBER
 SEQUENCE#                               NUMBER
 BYTES                                   NUMBER
 MEMBERS                                 NUMBER
 ARCHIVED                                VARCHAR2(3)
 STATUS                                  VARCHAR2(16)
 FIRST_CHANGE#                           NUMBER
 FIRST_TIME                              DATE
```

```
SQL> select group#, members, bytes, archived, status from v$log;

    GROUP#    MEMBERS      BYTES        ARC    STATUS
----------    ----------   ----------   ---    ----------------
         1          1      52428800     NO     CURRENT
         2          1      52428800     NO     INACTIVE
         3          1      52428800     NO     INACTIVE\
```

12) Adaugarea unui membru la un grup (adaugarea unui nou fisier de log):

```
SQL> alter database add logfile member 'E:\TEMP\log2.rdo' to group 1;
Database altered.
```

13) Resetarea unui grup dintr-un fisier de log(care nu este grup curent):

```
SQL> alter database clear logfile group 2 ;
Database altered.
```

14) Relocarea unui fiser de log cu baza de date online (trebuie ca fisierul sa nu fie in grupul curent si sa existe o copie in noua locatie):

```
SQL> alter database rename file 'E:\TEMP\log2.rdo' to 'E:\TEMP1\log2.rdo';
Database altered.
```

Observatii:

- ✓ Daca fisierul face parte din grupul curent, se genereaza eroarea:

```
ERROR at line 1:
ORA-01511: error in renaming log/data files
ORA-01621: cannot rename member of current log if database is open
```

ORA-00312: online log 1 thread 1:
' F:\ORA10G_DB\PRODUCT\10.2.0\ORADATA\UPB\REDO01.LOG'
ORA-00312: online log 1 thread 1: 'E:\TEMP\LOG2.RDO'
In acest caz se face un switch pe urmatorul grup, apoi se relocheaza fisierul respectiv.

- ✓ Daca fisierul nu a fost copiat deja in noua locatie se genereaza eroarea:

ERROR at line 1:
ORA-01511: error in renaming log/data files
ORA-01512: error renaming log file E:\TEMP\LOG2.RDO - new file
F:\temp1\log2.rdo not found
ORA-27041: unable to open file
OSD-04002: unable to open file
O/S-Error: (OS 2) The system cannot find the file specified.

15) Stergerea unui membru dintr-un grup :

- ➢ Se verifica starea fiserului care va fi sters :

```
SQL> select * from v$logfile;

GROUP#     STATUS     TYPE       MEMBER                                  IS_
-------    -------    -------    ------------------------------          ---
     3                ONLINE     C:\ORACLE_12C\ORADATA\BD\REDO03.LOG      NO
     2                ONLINE     C:\ORACLE_12C\ORADATA\BD\REDO02.LOG      NO
     1                ONLINE     C:\ORACLE_12C\ORADATA\BD\REDO01.LOG      NO
     1          INVALID  ONLINE     E:\TEMP1\LOG2.RDO                        NO
```

- ➢ Se sterge fisierul de log (trebuie sa nu fie in grupul curent) :

```
SQL> alter database drop logfile member 'E:\TEMP1\LOG2.RDO';
```

Observatii:

- ✓ Daca fisierul face parte dintr-un grup urmator nu se poate sterge si se genereaza eroarea:

ERROR at line 1:
ORA-00362: member is required to form a valid logfile in group 3
ORA-01517: log member: 'E:\temp\log2.rdo'

- ✓ Daca fisierul face parte din grupul curent, se genereaza eroarea:

ERROR at line 1:
ORA-01609: log 1 is the current log for thread 1 - cannot drop members

In acest caz se face un switch pe urmatorul grup, apoi se sterge membrul respectiv:

```
SQL> alter system switch logfile;
System altered.

SQL> alter database drop logfile member 'E:\TEMP\LOG2.RDO';
Database altered.
```

SQL> select * from v$logfile;
SQL> select * from v$logfile;

GROUP#	STATUS	TYPE	MEMBER	IS_
3		ONLINE	C:\ORACLE_12C\ORADATA\BD\REDO03.LOG	NO
2		ONLINE	C:\ORACLE_12C\ORADATA\BD\REDO02.LOG	NO
1		ONLINE	C:\ORACLE_12C\ORADATA\BD\REDO01.LOG	NO

16) Informatii legate de modul de lucru al instantei (modul arhiva sau nonarhiva):

SQL> desc v$instance

Name	Null?	Type
INSTANCE_NUMBER		NUMBER
INSTANCE_NAME		VARCHAR2(16)
HOST_NAME		VARCHAR2(64)
VERSION		VARCHAR2(17)
STARTUP_TIME		DATE
STATUS		VARCHAR2(12)
PARALLEL		VARCHAR2(3)
THREAD#		NUMBER
ARCHIVER		VARCHAR2(7)
LOG_SWITCH_WAIT		VARCHAR2(15)
LOGINS		VARCHAR2(10)
SHUTDOWN_PENDING		VARCHAR2(3)
DATABASE_STATUS		VARCHAR2(17)
INSTANCE_ROLE		VARCHAR2(18)
ACTIVE_STATE		VARCHAR2(9)
BLOCKED		VARCHAR2(3)

SQL> select instance_name, database_status, archiver from v$instance;

INSTANCE_NAME	DATABASE_STATUS	ARCHIVE
UPB	ACTIVE	STOPPED

17) Informatii despre istoricul fisierelor de log:

SQL> desc V$LOGHIST

Name	Null?	Type
THREAD#		NUMBER
SEQUENCE#		NUMBER
FIRST_CHANGE#		NUMBER
FIRST_TIME		DATE

SWITCH_CHANGE# NUMBER
CON_ID NUMBER

SQL> select * from V$LOGHIST;

THREAD#	SEQUENCE#	FIRST_CHANGE#	FIRST_TIM	SWITCH_CHANGE#	CON_ID
1	1	2233668	08-MAY-19	2251966	0
1	2	2251966	08-MAY-19	2257658	0
1	3	2257658	08-MAY-19	2257826	0
1	4	2257826	08-MAY-19	2260506	0
1	5	2260506	08-MAY-19	2370497	0
1	6	2370497	10-MAY-19	2385355	0
1	7	2385355	11-MAY-19	2393508	0
1	8	2393508	11-MAY-19	2401645	0
1	9	2401645	11-MAY-19	2417525	0
1	10	2417525	11-MAY-19	2426821	0
1	11	2426821	12-MAY-19	2441691	0